5. Exercise

Linear Models and Regression

Exercise Sheet 5

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Solve the following problems before the exercise class on December 11. Use R for all computations and all statistical tests if no output from R is given.

Problem 1

Alligator Airways operates two types of planes, one turboprop and one jet. The relation between the trip time (including boarding and de-boarding) and the distance of a trip is given by

$$\texttt{trip_time} = c_i + a_i \cdot \texttt{distance}$$

for $i \in \{\texttt{jet}, \texttt{turboprop}\}$. CEO Lion Tiger is interested in comparing $c_{\texttt{jet}}$ with $c_{\texttt{turboprop}}$ and $a_{\texttt{jet}}$ with $a_{\texttt{turboprop}}$. Thus, he considers the Gaussian linear model

$$\texttt{trip_time} = \gamma_0 + \gamma_1 \cdot \mathbf{1}\{\texttt{type} = \texttt{turboprop}\} + \gamma_2 \cdot \texttt{distance} + \gamma_3 \cdot \mathbf{1}\{\texttt{type} = \texttt{turboprop}\} \cdot \texttt{distance} + \varepsilon$$

with $\gamma_0, \gamma_1, \gamma_2, \gamma_3 \in \mathbb{R}$ and a centred normal random variable ε. He has 100 observations of the trip time in minutes (`trip_time`), the distance in kilometres (`distance`) and the type (factor `type` with levels `jet` and `turboprop`).
Use the level 0.01 and do not correct for multiple testing.

a) Formulate the null hypotheses and the alternatives to compare $c_{\texttt{jet}}$ with $c_{\texttt{turboprop}}$ and $a_{\texttt{jet}}$ with $a_{\texttt{turboprop}}$ in terms of $\gamma_0, \ldots, \gamma_3$.

b) Decide the test problems from a) and discuss the relations between $c_{\texttt{jet}}$ and $c_{\texttt{turboprop}}$ and between $a_{\texttt{jet}}$ and $a_{\texttt{turboprop}}$ using the following output in R:

```
Call:
lm(formula = trip_time ~ type * distance, data = flights)

Residuals:
     Min       1Q  Median      3Q      Max
-31.1022  -4.7392  0.9266  7.2582  24.5726

Coefficients:
                        Estimate  Std. Error  t value  Pr(>|t|)
(Intercept)            41.803610    4.736531    8.826  4.94e-14  ***
typeturboprop         -20.667462    6.777941   -3.049   0.00296  **
distance                0.074296    0.003593   20.679   < 2e-16  ***
typeturboprop:distance  0.050764    0.010767    4.715  8.19e-06  ***
--
```

#Recall

$$Y = \gamma_0 + \gamma_1 x_1^3 + \gamma_2 \log x_2 + \varepsilon \rightarrow \text{linear model}$$

$\downarrow$

#Note: linearity is about coefficients not covariance

$$Y = \gamma_0 e^{\gamma_1 x_1} \sqrt{\gamma_2 x_2} \rightarrow \text{not linear.}$$

#Question 1

$$\text{trip_time} = c_i + a_i \, \text{distance} \qquad i \in \{\text{jet}, \text{turboprop}\}$$

$\downarrow$

$$\text{trip_time} = \gamma_0 + \gamma_1 \mathbb{1}_{\{\text{type} = \text{turboprop}\}} + \gamma_2 \cdot \text{distance} + \gamma_3 \mathbb{1}_{\{\text{type} = \text{turboprop}\}} \cdot \text{distance} + \varepsilon$$

$\downarrow$

We have

$$\gamma_0 = c_{jet} \qquad \gamma_1 = c_{turboprop} - c_{jet}$$

$$\gamma_2 = a_{jet} \qquad \gamma_3 = a_{turboprop} - a_{jet}$$

a) Compare c_{turbo} & c_{jet} $\rightarrow$ $c_{turbo} = c_{jet}$

$c_{turbo} - c_{jet} = 0$

$\gamma_1 = 0$

$\therefore$ We test

$H_0: \gamma_1 = 0$ vs $H_1: \gamma_1 \neq 0$

$H_0: \gamma_3 = 0$ vs $H_2: \gamma_3 \neq 0$ $[a_{turbo} = a_{jet}]$

b) For the first test,
the p-value is $0.00296 < 0.01$

so we can reject H_0 at level 0.01

i.e., $c_{jet} \neq c_{turboprop}$

Since estimate of γ_1 is negative

$\hat{\gamma}_1 < 0$

then $c_{turboprop} - c_{jet} < 0$

i.e., $\boxed{c_{jet} > c_{turboprop}}$

For the 2nd test, the p-value is $8.19e^{-6} < 0.01$

So we can reject H_0 at level 0.01

i.e., $a_{turboprop} - a_{jet} \neq 0$.

since $\hat{\gamma}_3 > 0$

$\boxed{a_{turboprop} > a_{jet}}$

c) Trip time for type jet

$$\begin{aligned} \text{trip_time} &= \hat{\gamma}_0 + \hat{\gamma}_2 \cdot 350 \\ &= 41.80360 + 0.074296 \times 350 \\ &= 67.80721 \end{aligned}$$

for turboprop

$$\begin{aligned} \text{trip_time} &= \hat{\gamma}_0 + \hat{\gamma}_1 + \hat{\gamma}_3 \cdot 350 \\ &= 64.907144 \end{aligned}$$

d) We test

$H_0: \gamma_1 = \gamma_2 = \gamma_3 = 0$

against

$H_1: \gamma_i \neq 0$ for $i \in \{1,2,3\}$

p-value $< 2.2e-16$

Question 2

$$\text{Volume} = c \cdot \text{Girth}^{\alpha_1} \text{Height}^{\alpha_2}$$

$$\ln \text{Volume} = \ln c + \alpha_1 \ln \text{Girth} + \alpha_2 \ln \text{Height}$$

b) We can rewrite the linear model

$\alpha_1 = 2$ against $\alpha_2 = 1$

$$(\alpha_1 - 2) = 0 \quad \text{vs} \quad (\alpha_2 - 1) = 0$$

$$\boxed{\ln \text{volume} - 2\ln \text{Girth} - 1 \ln \text{Height}} \rightarrow \text{dependent.}$$

$$= \ln c + (\alpha_1 - 2) \ln \text{Girth} + (\alpha_2 - 1) \ln \text{Height} + \varepsilon$$

Since p-value $= 0.8464 > 0.01$

we cannot reject H_0

Question 3

$$\text{Volume} = g(\text{Girth}, \text{Height}; c, \alpha_1, \alpha_2)$$

$$g(G, H; c, \alpha_1, \alpha_2) = c \cdot G^{\alpha_1} H^{\alpha_2}$$

$$\frac{\partial g}{\partial c} = G^{\alpha_1} H^{\alpha_2}$$

$$\frac{\partial g}{\partial \alpha_1} = c \cdot H^{\alpha_2} \cdot \ln(\alpha_1) \cdot G^{\alpha_1}$$

$$\frac{\partial g}{\partial \alpha_2} = c \cdot G^{\alpha_1} \cdot \ln(\alpha_2) \cdot H^{\alpha_2}$$

The LSE (c, α_1, α_2) needs to satisfy

$$\sum_{i=1}^{n} \left(v_i - \hat{c}\, G_i^{\hat{\alpha}_1} H_i^{\hat{\alpha}_2} \right) \begin{pmatrix} G_i^{\hat{\alpha}_1} H_i^{\hat{\alpha}_2} \\ c \cdot H_i^{\hat{\alpha}_2} \cdot \ln(G_i) \cdot G_i^{\hat{\alpha}_1} \\ \hat{c} \cdot G_i \ln(H_i) \cdot H_i^{\hat{\alpha}_2} \end{pmatrix} = 0$$

c) $\ln \text{volume} = \ln C + \alpha_1 \ln G + \alpha_2 \ln H + \boxed{\varepsilon}$ additive noise

$\Leftrightarrow$ $\text{volume} = C \cdot G^{\alpha_1} H^{\alpha_2} \times \boxed{e^{\varepsilon}}$ exponential noise

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Question 4

$$g(t;\beta) = \frac{\beta_1}{1+\beta_2 \cdot e^{-\beta_3 t}}$$

$$\left(\frac{u}{v}\right)' = \frac{u'v - v'u}{v^2}$$

$$\frac{\partial g}{\partial \beta_1} = \frac{1}{1+\beta_2 \cdot e^{-\beta_3 t}}$$

$$\frac{\partial g}{\partial \beta_2} = \frac{0 - e^{-\beta_3 t} \cdot \beta_1}{(1+\beta_2 \cdot e^{-\beta_3 t})^2} = \frac{-\beta_1 e^{-\beta_3 t}}{(1+\beta_2 \cdot e^{-\beta_3 t})^2}$$

$$\frac{\partial g}{\partial \beta_3} = \frac{0 - \beta_1 \cdot \left[\beta_2 \cdot (-t) e^{-\beta_3 t}\right]}{(1+\beta_2 \cdot e^{-\beta_3 t})^2} = \frac{\beta_1 \beta_2 \cdot t \cdot e^{-\beta_3 t}}{(1+\beta_2 \cdot e^{-\beta_3 t})^2}$$

The LSE = MLE needs to satisfy

$$\sum_{i=1}^{n} \left(y_i - \frac{\hat{\beta}_1}{1+\hat{\beta}_2 e^{-\hat{\beta}_3 t}} \right) \begin{pmatrix} \frac{1}{1+\hat{\beta}_2 \cdot e^{-\hat{\beta}_3 t}} \\ \frac{-\hat{\beta}_1 e^{-\hat{\beta}_3 t}}{(1+\hat{\beta}_2 \cdot e^{-\hat{\beta}_3 t})^2} \\ \frac{\hat{\beta}_1 \hat{\beta}_2 \cdot t \cdot e^{-\hat{\beta}_3 t}}{(1+\hat{\beta}_2 \cdot e^{-\hat{\beta}_3 t})^2} \end{pmatrix}$$